Securing IoMT: Intrusion Detection to Improve Medical Data Security on Internet of Medical Things Using Machine Learning Techniques

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Abstract The Internet of Medical Things (IoMT) has made it possible for digital devices to collect, infer, and disseminate data related to health through the use of cloud computing. Securing data for use in health care has unique challenges. Various studies have been carried out with the aim of securing healthcare data. The best way to protect sensitive data is to encrypt them so that no one can decipher them. Conventional encryption methods are inapplicable to e-health data due to capacity, redundancy, and data size restrictions, particularly when patient data is transmitted across unsecured channels. Due to the inherent dangers of data loss and confidentiality breaches associated with data, patients may no longer be able to fully protect the privacy of their data contents. These security threats have been recognized by researchers, who have then proposed various methods of data encryption to fix the problem. As a result, the area of computer security is deeply concerned with finding solutions to the security and privacy issues associated with IoMT. This research presents an intrusion detection system (IDS) for IoMT that utilizes the machine learning techniques: Decision Tree (DT), Naive Bayes (NB), and K-Nearest Neighbor (KNN). Feature scaling using the minimum–maximum (min–max) normalization method was performed on the CIC IoMT 2024 dataset to prevent information leakage into the test data. The effectiveness of the output was then evaluated, ensuring that the scaling process was correctly implemented as the initial step of this work approach. Five types of assaults are identified in this dataset: DDoS, DoS, RECON, MQTT, and Spoofing. Principal component analysis (PCA) was used to reduce dimensionality in

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the subsequent stage. The suggested methods have a high detection rate of accuracy 98.2%, specificity of 97.6%, recall of 98.0%, and $F1$-score of 97.8%, which offer a viable option for protecting IoMT devices from attacks.

Keywords Machine learning · Medical data · IoT · Smart health · Internet of Medical Things (IoMT) · Network attacks

1 Introduction

The Internet of Medical Things (IoMT) is a subset of the Internet of Things (IoT) that encompasses medical devices that are connected to a variety of networks and are utilized for the purpose of monitoring health care [1]. The IoMT allows for hands-free healthcare monitoring with the integration of automation, interfacial sensors, and AI powered by machine learning. Through the use of remote access to data processing, transfer via a secure network, and medical device data gathering, the IoMT removes physical obstacles between patients and clinicians. IoMT technologies provide wireless monitoring of health parameters, which in turn reduces the need for needless hospital visits and the expenses associated with them. Many different types of medical technology are part of the IoMT, which includes things like wearables, POC devices used in healthcare institutions, and indoor personal real-time health monitoring [2]. Technologies that are already in existence are continuously being improved or altered by incorporating new sources of energy. For example, the IoT relies heavily on the most recent advancements in network technology, such as the transition from 1 to 5G networks. Concerned about potential privacy and security issues, this idea stimulates the interest of researchers, especially when considering scenarios with large bandwidth and frequency [3]. More base stations will be required to service the same area as the other wireless mechanism due to the short wavelength, which is expected to alter the infrastructure. The new layout has the potential to introduce additional attacks, like a proliferation of false base stations. Security of patient record data and hospital information is paramount in an IoMT-enabled infrastructure.

Managing security for the IoT takes three factors into account. Make sure you can identify the devices as being part of a network first. The second step is to specify the protocols that other programs, systems, and tools use for data exchange [4]. Finally, confirm that these IoMT devices haven't disrupted other network or organizational equipment in the event of an issue. The ability to diagnose, copy, and retrieve a vast amount of digital data is made possible by the accessibility of IoMT apps on a global scale. It frequently leads to the creation of counterfeit goods or their unlawful use in question [5]. Consequently, a lot of people have been working on methods to make IoMT applications more secure for medical data [6]. Consequently, encryption is a viable option for safeguarding medical images. The present encryption approaches have utilized the data encryption standard, the advanced encryption standard, and the Rivest–Shamir–Adleman (RSA) algorithms to tackle the problems of low-level efficiency while considering small data sizes and high redundancy [7]. Therefore, these approaches are difficult enough to oversee and ensure proper encryption of medical data inside the IoMT framework [8]. In order to make the most significant contribution, a new lightweight machine learning technique was presented. The privacy of patients' medical records was the driving force for the proposal of this method. Healthcare providers can improve patient care, save costs, and protect sensitive medical information by incorporating intrusion detection systems based on machine learning into smart health apps. When transmitted to healthcare organizations, the suggested method ensures that patients' personal information remains private and secure.

Here is the outline for the rest of the article: In Sect. 2, the relevant research is detailed. The methods and dataset are detailed in Sect. 3. The results are explained in depth in Sect. 4. Finally, Sect. 5 concludes the paper.

2 Related Work

Artificial intelligence (AI) has garnered attention from researchers and industries associated with biomedicine due to its ability to manage vast quantities of data, produce precise outcomes, and oversee processes to produce the most optimal output. AI is not new since robots are already used for decision-making and long-term illness prediction. Machines and algorithms now facilitate most everyday activities. Many healthcare organizations, including hospitals, have profited substantially from the IoMT [9]. However, new technologies provide challenges for businesses which use this innovation to fix the IoMT's security and privacy issues. Medical systems are vulnerable to one another when it comes to data and information availability, confidentiality, and integrity [10]. Security issues may impair confidentiality, risk integrity, and cause availability issues. Numerous research works have been conducted on the new technology known as IoMT. In [11], the author proposed a protocol and framework for an IoT patient monitoring system and advocated for its use to track patients' health data. Several communication methods, such as Bluetooth, RFID, Wi-Fi, and sensors, have been integrated into the medical nursing system (MNS) that is based on the IoT [12]. The authors of [13] suggested a system for patient monitoring that would use the patient's weight as an input. A different IoT solution was described in [14] to track and monitor autistic people by the use of sensors and signal collection from the brain. In order to transform antiquated hospitals into smart hospitals that can handle information in a more modern manner, a new IoT-based plan was proposed [15]. Another design to monitor elderly patients' health utilizing an end-to-end medical healthcare system method was introduced in [16], employing IoT as a back-end platform. An experienced healthcare system called Mycin (an AI-based system to identify bacteria causing different diseases) and RFID technology were used in an IoT-based approach for the electronic healthcare unit proposed by a researcher in [17].

In order to find the relationship that could be harmful, [18] has suggested a machine learning method in which the past network attacks; the technology may also pinpoint their exact locations. The research used four ML algorithms—C4.5, BayesNet, DT, and NB to predict the attackable target. The Bayesian network achieved the best overall accuracy (91.68%) among the tested methods. An intriguing proposal was proposed in [19] to educate laypeople about the condition via a bot. By connecting it to various smartphone sensors, this bot can provide a more flexible service through the IoM. In [20], the idea is emphasized that patients will greatly benefit from early disease prediction. Using the IoT, which aids patients in the field of remote healthcare systems, this can be accomplished.

3 Methodology

In this part, the suggested intrusion detection system (IDS) for detecting assaults on IoMT applications is discussed. A smartphone and its sensors send their data to a cloud storage service. Due to the data's susceptibility to attacks, data stored on the cloud cannot be considered secure. The cloud or transmission can be used to launch the attack. The CIC IoMT 2024 dataset was used for this study. This dataset was newly released and includes the most recent forms of attacks. Stage one of this research process involved normalizing the CIC dataset using the min–max idea and then scaling features to reduce data leakage. Afterward, PCA was used to reduce dimensionality. Following dataset collecting and loading, data preparation was the initial analysis undertaken. Eliminating outliers and unnecessary features is a crucial part of data preparation. In order to prepare the data for analysis, the Normalization approach utilizing the Min–max methodology was employed. PCA is a feature selection procedure that takes the min–max result as input. Following the reduction of the dataset, the NB, DT, and KNN classifiers are trained. Figure 1 depicts the structure of our suggested model.

3.1 Dataset Description

In the realm of healthcare cybersecurity, the CIC IoMT 2024 dataset (See Table 1) presents a simulated healthcare environment for researchers to investigate and develop security solutions for IoMT devices [21]. This dataset captures network traffic generated during cyberattacks perpetrated against a diverse testbed of 40 IoMT devices, including a mix of real and emulated ones. The captured traffic encompasses various protocols commonly used in healthcare settings, offering a realistic training ground for researchers. By leveraging machine learning on this rich dataset, researchers can develop models capable of identifying and potentially preventing various attack vectors, including denial-of-service assaults, reconnaissance attempts, and spoofing attacks targeting the MQTT protocol used for IoMT communication.

Ultimately, the CIC IoMT 2024 dataset serves as a valuable tool for researchers to analyze attack patterns, benchmark security solutions, and contribute to the ongoing

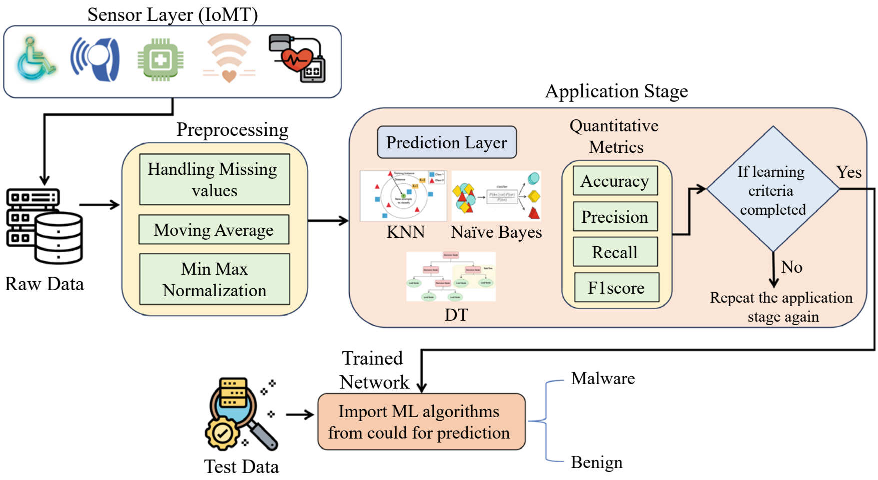


Fig. 1 Proposed IoMT-enabled smart model for intrusion detection using ML models

Table 1 CICIoMT2024 dataset's instance count for each class

Class	Category	Attack	Count
Benign	–	–	229,000
ATTACK	Spoofing	APR	16,600
	Recon	OS Scan	22,666
	MQTT	Dos P Food	50,800
	DoS	DoS UDP	704,503
	DDoS	DDoS UDP	1,998,026

effort of safeguarding patient privacy and device functionality within the healthcare domain.

3.2 Data Preprocessing and Normalization

Data preparation is an important part of machine learning since it helps to remove errors from the dataset [8]. The suggested process begins with this step. According to projections, it will transform the raw data from network IoT attacks into a usable format for additional study. Feature scaling with the goal of having all attribute values on the same scale is known as normalization. Standardized moment, z-score, and min–max normalization are among the many normalizing methods available [5]. In this research, we utilized the min–max normalization technique.

3.3 Principal Component Analysis

The PCA is the first and most well-known method in multivariate statistical analysis, and it is also one of the most used unsupervised methods for feature selection. Using PCA helps lower the dimensionality while preserving the information about the important qualities in the dataset [5]. Through the use of orthogonal linear combinations with a substantial variation, it lowers the variable count. When it comes to categorization, PCA also chooses the most relevant features from the dataset. In this study, we employ PCA to reduce the dimensionality of the CIC IoMT dataset by compressing the attribute space using components. The most dissimilar primary characteristics are linearly mixed to form the y_1 principal component.

This first component's values are displayed in Eq. (1):

$$Y_1 = Lb_1. \tag{1}$$

As stated in Eq. (1), M represents the number of samples, L represents the matrix of observations with a mean of zero, and b_1 represents the vector that has the highest variance (y):

$$\frac{1}{M \cdot y_1^1 y_1} = \frac{1}{M \cdot b_1^1 U^1 U_{a_1}} = b_1 K b_1. \tag{2}$$

K represents the post-observational matrix of covariances and variances. Recognizing the constraint $b_1 = 1$ using the Lagrange multiplier theory is necessary to discover the solution to Eq. (2).

$$Z = b_1 K \cdot b - v(b_1 \cdot b_1 - 1). \tag{3}$$

Maximizing Eq. 3 entails deriving it in terms of its constituents b_1 until zero:

$$\partial Z / (\partial b_1) = 2Db_1 - 2vb_1 = 0. \tag{4}$$

Equation 4 gives in $Db\_1 = vb_1$, where b is an eigenvector of D, and v is the eigenvalue.

3.4 K-Nearest Neighbor

KNN identifies the k closest data points to a given input and assigns the most common classification among these neighbors. When deployed, KNN continuously monitors IoMT network traffic, comparing each new data point to its nearest neighbors to detect and flag potential intrusions, thereby enhancing the security of IoMT systems.

In this study, the ten components' properties chosen by principal component analysis (PCA) were classified using the KNN.

Algorithm 1: **K-nearest model**

Input: CIC dataset
    pre-process
    centroid of the neighbor to the $\rightarrow K^{th}$ power
Output: k and P_k
  for i = 1 until the dataset invades
    for k = 1 the cluster dataset's centroid
    S_k = Find the geometric distance between points k and i
    E = $\sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2}$
    end for
Sort S_k distances
Pk $\rightarrow$ Calculate probability of k
    end for
Pick the one with the best probability
return: highest prob P_k

3.5 Naïve Bayes

The Naïve Bayes algorithm is one of several algorithm families that are founded on the naïve Bayes theorem. They both agree that every set of predictions stands alone. In addition, it notes that features are related to the prediction in a similar but independent way. In addition, it is noted that the features contribute to the prediction in similar yet independent ways, which enhances the model's interpretability and performance. This characteristic has been further optimised for application in data mining tasks [16]. By combining these probabilities, the method uses Bayes' theorem to find the most likely class for new data points.

Algorithm 2: NB model

Input: training of CIC dataset D
dataset d = $\{d_1, d_2, d_3 \ldots d_n\}$
S is the chosen number of clusters
Output: Set of s number of cluster
Code: Read the dataset D
repeat: Calculate posterior probability
P(Class|Data) = $\frac{\text{P(Data|Class)} * \text{P(Class)}}{\text{P(Data)}}$
Determine the average and standard deviation of d
Calculate probability for $d_1, d_2, d_3 \ldots d_n$
end

3.6 Decision Tree

Internet of Medical Things (IoMT) intrusion detection can be accomplished with the use of a decision tree by using parameters like packet size, protocol type, and connection time to categorize network traffic as benign or malicious. The decision tree learns to partition data at internal nodes using these features through training on datasets such as NSL-KDD or UNSW-NB15. The branches of the tree indicate decision rules, and the leaf nodes show the classification outcome. Once trained, the model enhances the security of IoMT systems by monitoring network traffic in real-time and applying its learnt rules to detect and flag potential attacks. The entropy of each node is calculated as:

$$\text{Entropy } (S) = \sum (j = 1)\hat{}c - p_j \log_2 \big(p_j\big), \tag{5}$$

$$P(\text{yes}) = \frac{P(\text{yes})}{P(\text{yes}) + P(\text{no})}, \text{ for Yes component of node } A. \tag{6}$$

An entropy value of 0 indicates a perfectly pure subtree, whereas an entropy value of 1 indicates the most impure subtree.

Algorithm 3: Decision tree

Input: D, where D → set of datasets
Output: decision tree
Procedure: build tree
  repeat
    Max gain = 0
    Split the B = null
    S → the entropy (Attributes)
  for all the attributes of b in D do
    gain = Information increase (b, S)
    if gain is greater than max gain
      max gain = actual gain
      split B = b
    end if
  end for
end procedure

4 Methodology

The results and discussion of the experimental analysis are presented. There were five distinct kinds of attacks that CIC launched. The following categories apply: DDoS, DoS, Recon, MQTT, and spoofing. One-half of the dataset was utilized for model training, while the other half was utilized for model testing.

Using the CIC dataset, the intrusion detection system was able to successfully recognize and categorize several sorts of attacks. Denial-of-service (DoS) assaults were identified by the system as an attempt to limit the services that legitimate users could access. Neptune, Smurf, Teardrop, and SYN flooding are all instances of denial-of-service assaults. In addition, the system proved that it could identify Recon assaults, in which an attacker takes over local machines by taking advantage of security holes. There are a variety of U2R attacks, such as SQL injections, buffer overflows, rootkit installations, and eavesdropping. Moreover, MQTT assaults, in which unauthorized users try to access remote machines, were successfully detected by the intrusion detection system. Tools like as Nmap, port scanning, ping sweeps, and Satanic network mapping are all examples of probe assaults. This shows how strong the intrusion detection system is because it was able to identify and categorize these kinds of attacks. The system can respond quickly and assist reduce security risks by correctly identifying and classifying various types of threats. The CIC dataset results show that the system can handle many different types of intrusion attempts, which is great for protecting the integrity and security of IoMT devices and systems. Figure 2 displays the relevant performance details of the classifiers used for IoMT intrusion detection systems, including their accuracy, recall, specificity, and $F1$-score. In the context of IoMT, the recall values of the classifiers used for intrusion detection systems (IDS) show how well they identify positive case. Naive Bayes outperformed the other classifiers (see Figs. 3 and 4) with an impressive accuracy of 98.2%, specificity of 97.6%, recall of 98.0%, and $F1$-score of 97.8%, pertaining to IoMT system intrusions. With an accuracy of 93.0%, specificity of 92.8%, recall of 95.7%, and $F1$-score of 94.2%, K-Nearest Neighbors (KNNs) correctly identified, which is lower than expected. These findings demonstrate that NB has better recall performance than its competitors, demonstrating its promise as a dependable and strong classifier for IoMT intrusion detection systems. To thoroughly assess the classifiers' performance and guarantee their appropriateness for real-world IoMT scenarios, additional metrics and parameters must be taken into account in addition to recall. The effectiveness of the intrusion detection system (IDS) in precisely recognizing different cyber intrusion types within the IoMT environment is demonstrated by the study of findings obtained from deploying the CIC IoMT dataset (see Table 2). The system exhibits a thorough comprehension of several attack types, such as DDoS, DoS, Recon, MQTT, and spoofing.

However, some drawbacks emerge, like the possible difficulties in managing new and advanced cyber threats that are not sufficiently represented in the training dataset. The various classifiers' unique performances, especially NB better recall performance, highlight how important parameter selection is in determining the system's capacity for prediction. Moreover, the efficacy of the system can be improved by integrating sophisticated machine learning methodologies, such as ensemble approaches and deep learning, to strengthen its capacity to identify previously undetected attack patterns.

These findings demonstrate the system's potential use in enhancing the security posture of linked medical equipment, safeguarding private patient information,

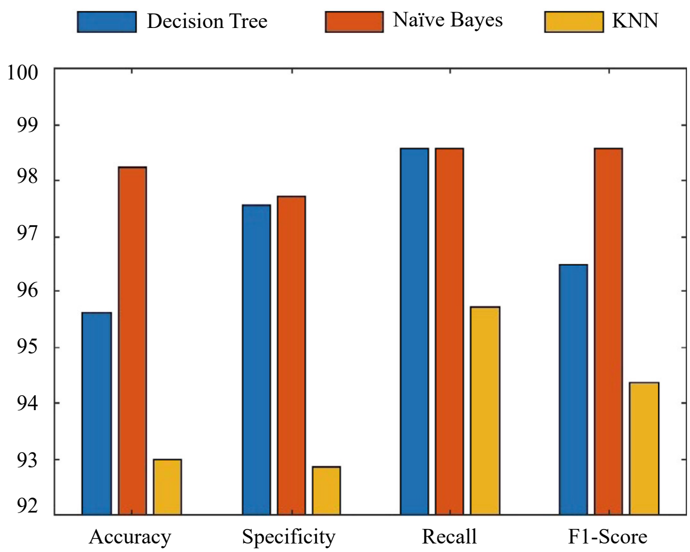
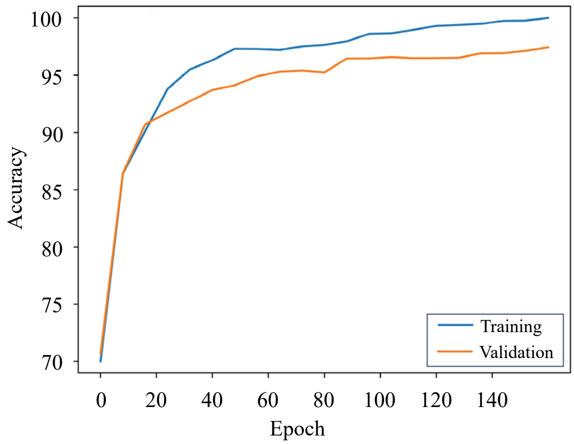


Fig. 2 Comparison results with accuracy, precision, recall, $F1$-score

Fig. 3 Accuracy versus loss for ML-IDS in IoMT dataset



and facilitating the seamless incorporation of cutting-edge telemedicine and telehealth services into the dynamic healthcare sector. Because cyber threats are always evolving, it is crucial to update and upgrade the IDS often to keep it powerful and reliable in protecting the IoMT ecosystem.

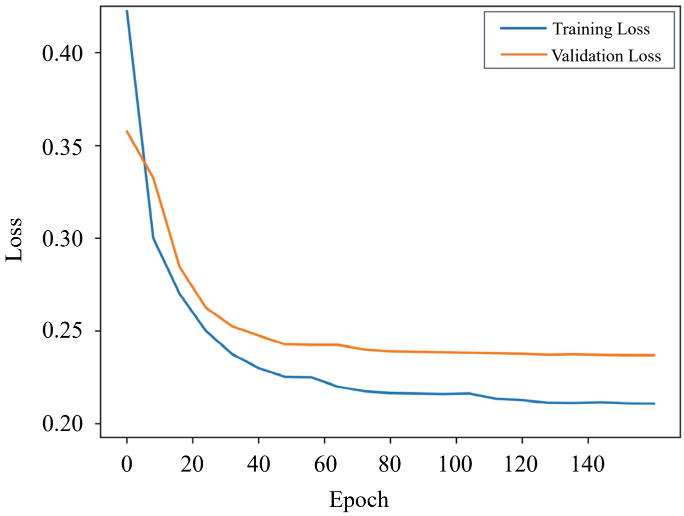


Fig. 4 Epoch versus loss for ML-IDS in IoMT dataset

Table 2 Comparison of the proposed system with state-of-the-art approaches

Reference	Method	Dataset	Accuracy	Precision	Recall	$F1$-score
Zhong et al. [5]	ANN	UNSWNB-15	86.53	×	×	×
Thamilarasu et al. [11]	XG Boost	Network Traffic Data	93.6	×	93.7	64.0
Kondaka et al. [13]	Random Forest	NSL-KDD	94.27	92.18	92.29	×
Saba et al. [14]	SVM	KDDCUP-99	94	98	96	×
Anitha et al. [18]	KNN	DDoS	91.5	×	×	×
X Wu et al. [19]	ANN, Cat Boost	CICIDS	95.6	93.4	92.9	90.2
This work	NB	CICIoMT-24	98.2	97.6	98.0	97.8

5 Conclusion

In this work, we explore the enormous potential of the IoMT to improve health care for those who require continuous medical care or monitoring. Integrating state-of-the-art health app platforms with machine learning intrusion detection systems is essential for improving therapy delivery while protecting patient privacy and sensitive medical data. In the realm of computer security, ML-IDSs have become increasingly popular due to their ability to detect threats automatically. It is possible that we might employ machine learning and optimization of intrusion detection techniques to fortify the security architecture surrounding medical IoT devices. As a result, this will finally ensure the health and safety of patients in our society, which is becoming increasingly interconnected.

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